# Categorical Spectralism: A Functorial Framework for Spectral Complexity and a Conjecture on Substrate-Independent Categorical Equivalence

Charles O'Connor

Monash University / Turner Institute for Brain and Mental Health

**Abstract**

Scientific classification has always been provisional. At every major theoretical advance, from Newtonian mechanics to general relativity, from classical particle physics to quantum field theory,categorical labels that appeared definitive were revealed to be approximations of a richer underlying structure. We introduce *Categorical Spectralism* (CS), a mathematical framework that formalises this observation. CS proposes that for any system with inaccessible internal structure, the appropriate mathematical object is not a categorical label but a spectral density distribution between physically grounded absolute boundaries, a continuous, substrate-independent signature carrying strictly more information than any binary classification. The framework is structured across three abstraction layers: a category of undefined objects $\mathcal{U}$ whose identity is exhausted by internal state transitions; a functor $F\colon \mathcal{U} \to \mathbf{Hilb}$ mapping morphic structure into a Hilbert space; and a spectral analysis of the resulting operator whose eigenvalue distribution constitutes the system's spectral signature. From this signature we derive the Spectral Complexity Index (SCI), a scalar functional on the spectrum computable in polynomial time from empirical spectral data and invariant under unitary equivalence. We instantiate the framework on the gravitational wave event GW150914, yielding $\mathrm{SCI}(\mathrm{GW150914}) = 3.14 \pm 0.08$ from published LIGO parameter estimates. We propose SCI as a theoretically grounded measure of conscious complexity, advancing a falsifiable prediction on the ordering of consciousness states. The central conjecture—that SCI equivalence implies categorical equivalence regardless of substrate—is stated precisely with a structured proof program and explicit falsification conditions.

© 2026 SpectralCo / Charles O'Connor. All rights reserved. Draft preprint — do not reproduce without attribution. spectralco.github.io

# Contents


**1 Introduction** **3**

**2 Related Work** **5**

2.1 Noncommutative Geometry and Spectral Triples . . . . . 5
2.2 Category-Theoretic Approaches to Consciousness . . . . . 5
2.3 Higher Category Theory and Topos-Theoretic Approaches . . . . . 6
2.4 Integrated Information Theory . . . . . 7
2.5 The Free Energy Principle . . . . . 7
2.6 Spectral Methods in Applied Mathematics . . . . . 8

**3 The Framework** **8**

3.1 Layer 1: The Category of Undefined Objects . . . . . 9
3.2 Layer 2: The Representation Layer . . . . . 10
3.3 Layer 3: The Spectral Layer . . . . . 12
3.4 The Spectral Complexity Index . . . . . 13

**4 Instantiation: Gravitational Wave Identification** **15**

4.1 Simple Instantiation: The Resonant Black Box . . . . . 15
4.2 Constructing $\mathcal{U}$ for the Gravitational System . . . . . 16
4.3 The Functor $F$ for the Gravitational System . . . . . 17
4.4 The Spectral Signature: Quasinormal Modes . . . . . 18
4.5 SCI Computation for GW150914 . . . . . 18
4.6 Interpretation and Predictions . . . . . 19

**5 Proposed Application: Consciousness** **20**

5.1 Why Consciousness Satisfies theorem 3.1 . . . . . 20
5.2 Constructing $\mathcal{U}$ for Consciousness . . . . . 20
5.3 The Functor $F$ for Consciousness . . . . . 21
5.4 The Spectral Signature: Connectome Harmonics . . . . . 21
5.5 Proposed SCI for Consciousness . . . . . 22
5.6 Theoretical Advantages of SCI over Existing Measures . . . . . 22
5.7 Open Problems for the Consciousness Application . . . . . 23

**6 The Central Conjecture** **23**

6.1 Preliminary Definitions . . . . . 23
6.2 The Central Conjecture . . . . . 24
6.3 What a Proof Requires . . . . . 25
6.4 Falsification Conditions . . . . . 26
6.5 Implications if True . . . . . 26

6.6 Relationship to the Open Problems of section 4 . . . . . 27

**7 Discussion** **27**

7.1 CS as Mathematical Structuralism . . . . . 27

7.2 CS and the Philosophy of Scientific Measurement . . . . . 27

7.3 Objections and Responses . . . . . 28

7.4 The Open Mathematical Programme . . . . . 29

7.5 Scope and Limits . . . . . 29

7.6 Relationship to the Broader Mathematical Landscape . . . . . 30

**8 Conclusion** **30**

# 1 Introduction

The classification of systems with inaccessible internal structure is a foundational problem in mathematics and the sciences. At every major theoretical advance,from Newtonian mechanics to general relativity, from classical particle physics to quantum field theory,categorical labels that appeared definitive were revealed to be approximations of a richer underlying structure. We argue that this pattern is not incidental but structural: classification is always provisional precisely because the objects being classified have inaccessible internal structure. The appropriate mathematical response is not to replace the labels but to supplement them with a richer object that carries strictly more information.

In each case the progression follows the same structure. A categorical label,-simultaneous events, particle, force,-functions effectively within a bounded domain of observation. At the boundary of that domain, the label breaks: it either fails to apply, applies ambiguously, or applies to objects that are empirically indistinguishable yet categorically distinct. Consider the degradation of simultaneity under general relativity: within Newtonian mechanics, the category of simultaneous events is absolute, observer-independent, and operationally clean. At relativistic velocities the category does not merely become imprecise,it becomes observer-dependent in a mathematically precise way, and no refinement of the label recovers its absoluteness. The category was not wrong; it was a limiting case of a continuous geometric object, the metric tensor, valid only within a restricted domain of the parameter space [Einstein, 1905, 1915]. The same degradation occurs for the category of particle under quantum field theory. Within classical mechanics a particle is a primitive,a localised, discrete object with definite position and momentum. At quantum scales the category first becomes probabilistic, then dissolves entirely into an excitation of a quantum field,a distribution over modes, not a localised object at all [Dirac, 1927, Weinberg, 1995]. Within the domains CS applies to,-systems with inaccessible internal structure,classificatory labels degrade in this direction: the domain of validity of the label shrinks relative to the full parameter space of the theory, and the continuous mathematical object underneath it expands to fill the space the label could not. We argue that this degradation is not a failure of scientific method but a structural feature of how mathematics encounters physical reality. The appropriate response is not to wait for the next theoretical advance to dissolve today's categories, but to begin with the continuous object and derive the label as a limiting case.

This paper introduces Categorical Spectralism (CS), a mathematical framework that formalises this response. CS proposes that for any system with inaccessible internal structure, the appropriate mathematical object is not a categorical label but a spectral density distribution between physically grounded absolute boundaries,a continuous, substrate-independent signature carrying strictly more information than any binary classification. The framework is structured across three abstraction layers: a category of undefined

objects whose identity is exhausted by their internal state transitions; a functor mapping this category into a complete Hilbert space; and a spectral analysis of the resulting operator whose eigenvalue distribution constitutes the system's spectral signature. From this signature we derive the Spectral Complexity Index (SCI), a scalar functional on the spectrum that enables direct cross-domain comparison of structurally distinct systems.

The framework connects to and extends several existing programmes. Connes' spectral triples [Connes, 1994] are the closest mathematical ancestor: CS extends the spectral triple construction by treating the algebra as undefined and recovering algebraic structure from morphic information, making the framework applicable to systems where the algebraic structure is empirically inaccessible. Tsuchiya and Saigo [Tsuchiya et al., 2016, Tsuchiya & Saigo, 2021] independently identified the need for a functor between the category of phenomenal structures and a mathematical space capable of grounding them; CS provides precisely this functor and generalises the construction beyond consciousness science. Integrated Information Theory [Tononi, 2004] and the Free Energy Principle [Friston, 2010] each provide domain-specific complexity measures; CS proposes a substrate-independent unification whose precise relationship to both frameworks is stated as an open problem.

The paper makes the following contributions. First, a formal three-layer mathematical framework for spectral complexity in systems with inaccessible internal structure. Second, a formally defined Spectral Complexity Index with stated axioms and partial proofs. Third, a complete instantiation of the framework applied to gravitational wave identification via LIGO data, yielding an exact SCI computation for GW150914. Fourth, a proposed application to consciousness science in which SCI is advanced as a theoretically grounded replacement for existing complexity measures, with empirical validation deferred to subsequent work. Fifth, a precisely stated central conjecture,-that spectral complexity equivalence implies categorical equivalence regardless of substrate,together with a statement of what a proof requires and the conditions under which it is falsified.

We are explicit about the status of each claim. The three-layer framework is formally constructed. Functoriality of the proposed functor is conjectured with a proof sketch provided; full proof is an open problem. The SCI axioms of non-negativity and unitary invariance are proven; monotonicity and boundedness are stated as open propositions. The central conjecture is stated as a conjecture. The consciousness application is theoretical; empirical validation is explicitly deferred. This transparency is deliberate,-the paper's primary contribution is a precisely stated mathematical programme, not a completed theory.

## 2 Related Work

Categorical Spectralism sits at the intersection of three mathematical traditions and two empirical programmes. This section positions CS precisely within each, defining all necessary mathematical objects before stating the relationship, and distinguishing where CS extends existing work, where it diverges, and where the relationship remains an open problem.

### 2.1 Noncommutative Geometry and Spectral Triples

We begin with the necessary definitions. A $C^*$-algebra is a Banach algebra $A$ equipped with an involution $*$ satisfying the $C^*$-identity $\|a^*a\| = \|a\|^2$ for all $a \in A$. A *spectral triple*, introduced by Connes [Connes, 1994], is a triple $(A, \mathcal{H}, D)$ consisting of a $C^*$-algebra $A$ represented on a Hilbert space $\mathcal{H}$ by bounded operators, together with a self-adjoint operator $D$,-the Dirac operator,-with compact resolvent. The spectral properties of $D$ encode geometric information about the underlying space. In the commutative case, Gelfand duality [Gelfand & Naimark, 1943] makes this precise: a commutative $C^*$-algebra $A$ is isomorphic to the algebra $C(X)$ of continuous functions on a compact Hausdorff space $X$, and the spectral properties of $D$ recover the metric geometry of $X$.

CS extends this construction in one critical direction. In the spectral triple, the algebra $A$ is assumed known. For systems with inaccessible internal structure, this assumption fails precisely because the algebra governing the system's internal dynamics is what cannot be accessed. CS removes this assumption entirely. Rather than beginning with $A$ and deriving spectral properties, CS begins with the morphic structure of observable state transitions and recovers both the operator and its spectral properties from that structure. The algebra, where it exists, is recovered as a consequence rather than stipulated as a premise. Whether the CS functor construction is a generalisation of spectral triples in a precise categorical sense, or an independent construction that subsumes them, is an open problem stated explicitly as part of the CS programme.

### 2.2 Category-Theoretic Approaches to Consciousness

Category theory, introduced by Eilenberg and Mac Lane [Eilenberg & Mac Lane, 1945], provides a mathematical language for structural relationships between mathematical objects. A category $\mathcal{C}$ consists of objects, morphisms between objects, identity morphisms, and a composition law satisfying associativity and unit laws. A functor $F\colon \mathcal{C} \to \mathcal{D}$ between two categories preserves this structure: it maps objects to objects, morphisms to morphisms, identities to identities, and composition to composition. A natural transformation $\eta\colon F \Rightarrow G$ between two functors assigns to each object $X$ a morphism $\eta_X\colon F(X) \to G(X)$ in a way compatible with the functors' action on morphisms.

Tsuchiya and Saigo [Tsuchiya et al., 2016, Tsuchiya & Saigo, 2021] applied this language to consciousness science, arguing that phenomenal structures,-the contents and levels of conscious experience,-form a category and that their relationships are best understood functorially. Their programme identified a precise mathematical gap: a functor was needed between the category of phenomenal structures and a mathematical space capable of grounding them rigorously and empirically. This functor was called for explicitly but not constructed.

CS provides precisely this functor. The category $\mathcal{U}$ of undefined objects, with phenomenal states as objects and experiential transitions as morphisms, maps under $F$ to a Hilbert space whose spectral properties ground the phenomenal structure without requiring a prior definition of consciousness. CS generalises the construction beyond consciousness,-the same functor applies to any system with inaccessible internal structure,-but the consciousness application remains the primary empirical motivation and the domain in which empirical validation is proposed.

### 2.3 Higher Category Theory and Topos-Theoretic Approaches

We introduce the necessary definitions. An $\infty$-category, or $(\infty, 1)$-category, is a generalisation of an ordinary category in which morphisms between objects may themselves have morphisms between them,-homotopies,-and homotopies between homotopies, to all higher levels. Lurie's Higher Topos Theory [Lurie, 2009] provides the foundational framework for $\infty$-categories, establishing the existence of an $\infty$-categorical analogue of sheaves and geometric morphisms. A topos is a category with the structural properties of the category of sets. Topos-theoretic approaches to quantum mechanics [Heunen & Landsman, 2009, Döring & Isham, 2011] use the internal logic of a topos to represent the state space of a quantum system without committing to a classical Boolean logic.

CS is related to but distinct from these approaches. The category $\mathcal{U}$ in CS is an ordinary category, not an $\infty$-category, and the functor $F$ targets **Hilb** rather than a topos. The key divergence is that topos-theoretic approaches to quantum mechanics typically begin with a known algebra of observables and reconstruct the state space, whereas CS begins with observable morphisms and reconstructs both the algebra and the state space. Whether the CS construction can be lifted to an $\infty$-categorical setting is an open question that the CS programme explicitly invites. Such a lifting would connect CS directly to Lurie's spectral algebraic geometry [Lurie, 2018] and could provide the formal machinery needed to prove the central conjecture.

### 2.4 Integrated Information Theory

Integrated Information Theory [Tononi, 2004, Oizumi et al., 2014] proposes that consciousness is identical to integrated information, measured by the scalar $\Phi$,-the amount of information generated by a system above and beyond its parts. To define $\Phi$ precisely: given a system $S$ in state $s$, one computes the cause-effect structure of $S$ by evaluating, for every possible partition of $S$ into two subsystems, the distance between the probability distribution specified by the whole system and the product distribution of the partitioned subsystems. $\Phi$ is defined as the minimum such distance over all partitions,-the minimum information partition (MIP).

IIT is substrate-independent in principle and produces a continuous scalar measure, both properties CS shares. The critical structural difference lies in the scope condition. IIT requires full enumeration of the system: its elements, their states, and their complete causal relationships must all be specified to compute $\Phi$. For systems with inaccessible internal structure,-the domain CS is defined for,-full enumeration is unavailable by definition, and $\Phi$ cannot be computed. Computing $\Phi$ requires iterating over all possible partitions of the system, a combinatorial search that grows super-exponentially with system size and is computationally intractable for systems beyond a small number of elements, as acknowledged by the framework's own authors [Oizumi et al., 2014]. SCI, by contrast, is computable in polynomial time from spectral data via standard eigendecomposition algorithms.

Whether SCI subsumes $\Phi$ as a special case,-that is, whether $\mathrm{SCI} = f(\Phi)$ for some function $f$ when full internal enumeration is available,-is an open problem stated explicitly as part of the CS research programme.

### 2.5 The Free Energy Principle

The Free Energy Principle [Friston, 2010] models biological systems as minimising variational free energy,-formally, an upper bound on the surprise of sensory observations given an internal generative model. The variational free energy $\mathcal{F}$ is defined as:

$$\mathcal{F} = \mathbb{E}_q\Big[\ln q(x) - \ln p(o, x)\Big],$$

where $q(x)$ is the approximate posterior over hidden states $x$, $p(o, x)$ is the generative model over observations $o$ and hidden states, and the expectation is taken under $q$.

FEP is a process theory: it describes the dynamics by which systems change over time and converge to attractors in their state space. CS is not a process theory. It characterises the spectral signature of a system's state space,-the structure of the attractor itself,-rather than the dynamics of convergence. The relationship between CS and FEP is therefore

complementary rather than competitive. A natural conjecture, deferred to future work, is that systems minimising free energy under FEP converge to attractors whose spectral signatures are invariant under the CS functor,-that is, that free energy minimisation preserves SCI at equilibrium.

### 2.6 Spectral Methods in Applied Mathematics

We define the relevant objects. The *graph Laplacian* of a weighted undirected graph $G = (V, E, w)$ is the matrix $L = D - A$, where $A$ is the weighted adjacency matrix with $A_{ij} = w(i, j)$ and $D$ is the diagonal degree matrix with $D_{ii} = \sum_j w(i, j)$. The eigenvalues of $L$,-the graph spectrum,-characterise connectivity, diffusion, and clustering properties of the graph without requiring categorical labels for nodes or edges [Chung, 1997]. The *Koopman operator*, introduced by Koopman [Koopman, 1931], is defined for a dynamical system with state space $X$ and evolution map $T\colon X \to X$ as the operator $K$ acting on the space of observables $f\colon X \to \mathbb{R}$ by $Kf = f \circ T$. The key property is that $K$ is a linear operator on an infinite-dimensional function space even when $T$ is nonlinear on the finite-dimensional state space $X$ [Mezic, 2005].

Graph spectral methods provide direct empirical precedent for Layer 3 of CS. CS situates these methods within a categorical framework: the spectral computation of the graph Laplacian is not new, but its derivation from a functor applied to a category of undefined objects, and its elevation to a substrate-independent complexity measure via SCI, are the contributions CS makes over existing graph spectral methods.

*Remark* 2.1 (CS as a generalisation of Koopman theory). CS is a strict generalisation of Koopman operator theory. When applied to a measure-preserving dynamical system $(X, \mu, T)$ with known state space and evolution map, the CS functor $F\colon \mathcal{U} \to \mathbf{Hilb}$ reduces to the Koopman operator $K\colon L^2(X, \mu) \to L^2(X, \mu)$ defined by $Kf = f \circ T$,-the functoriality conditions are satisfied by construction, since iteration of $T$ corresponds exactly to composition of Koopman operators. When the state space and evolution map are inaccessible, the CS construction proceeds from observable morphisms alone, which is precisely the regime Koopman theory cannot address. The extension of CS to dissipative systems with unbounded Koopman operators remains an open problem.

## 3 The Framework

We construct Categorical Spectralism across three abstraction layers. Each layer is formally defined and its relationship to the adjacent layers is made precise. The framework is built from the ground up: no prior definition of the system under study is assumed beyond the existence of observable state transitions.

Throughout this section we adopt the following scope condition, which defines the domain

of applicability of CS.

**Definition 3.1** (System with inaccessible internal structure)**.** A system $S$ has *inaccessible internal structure* if its internal states cannot be fully enumerated or directly observed. Inaccessibility is the criterion of entry into the CS framework,-it is not a property attributed to a prior object but the condition under which the framework applies. A system $S$ is *accessible* if its internal structure can be fully specified; CS does not apply to accessible systems and does not claim to.

Inaccessibility may arise in principle,-as in the interior of a black hole, which is causally disconnected from external observers,-or in practice,-as in the full synaptic state of a biological neural system, whose enumeration exceeds current and foreseeable observational capacity. CS applies to both cases without distinction.

### 3.1 Layer 1: The Category of Undefined Objects

**Definition 3.2** (Category of undefined objects)**.** Let $S$ be a system with inaccessible internal structure. The *category of undefined objects* $\mathcal{U}$ associated to $S$ is defined as follows:

(i) **Objects.** The objects of $\mathcal{U}$ are abstract states of $S$,-configurations of $S$ that are observationally distinguishable without requiring a definition of what $S$ fundamentally is. We denote the collection of objects $\mathrm{Ob}(\mathcal{U})$.

(ii) **Morphisms.** For any two objects $O, O' \in \mathrm{Ob}(\mathcal{U})$, a morphism $f\colon O \to O'$ is an observable internal state transition of $S$ from state $O$ to state $O'$. Morphisms represent transitions *within* $S$,-they are not relationships between $S$ and any external system. The collection of morphisms from $O$ to $O'$ is denoted $\mathrm{Hom}_{\mathcal{U}}(O, O')$.

(iii) **Identity morphisms.** For each object $O \in \mathrm{Ob}(\mathcal{U})$, the identity morphism $\mathrm{id}_O\colon O \to O$ is the trivial transition,-$S$ remaining in state $O$. This is always observable: the absence of transition is itself an observable.

(iv) **Composition.** For morphisms $f\colon O \to O'$ and $g\colon O' \to O''$, their composition $g \circ f\colon O \to O''$ is the transition from $O$ to $O''$ via the intermediate state $O'$. Composition is associative: for $h\colon O'' \to O'''$, we have $h \circ (g \circ f) = (h \circ g) \circ f$ since both represent the transition $O \to O'''$ through the same sequence of intermediate states.

(v) **Unit laws.** $\mathrm{id}_{O'} \circ f = f$ and $f \circ \mathrm{id}_O = f$ for all $f\colon O \to O'$, since composing a transition with a trivial transition leaves the transition unchanged.

**Proposition 3.3.** *For any system $S$ with inaccessible internal structure, theorem 3.2 yields a well-defined small category $\mathcal{U}$, provided that the collection of observable states of $S$ is a set.*

*Proof.* The five axioms of a category,-objects, morphisms, identity, associativity, and unit laws,-are verified directly by clauses (i),(v) of theorem 3.2. Smallness requires that $\mathrm{Ob}(\mathcal{U})$ is a set rather than a proper class, which holds whenever the observable states of $S$ are indexed by a set. For all physical systems considered in this paper,-gravitational wave systems characterised by finitely many quasinormal modes, conscious systems characterised by finitely many observable states,-this condition holds. □

*Remark* 3.4 (Self-containedness of $\mathcal{U}$). $\mathcal{U}$ is self-contained: it encodes the morphic structure of a single system $S$ and contains no inter-system relationships. Comparison between two distinct systems $S$ and $S'$,-each with their own category $\mathcal{U}$ and $\mathcal{U}'$,-does not occur at the level of $\mathcal{U}$ but at the level of the functor $F$, where the spectral images of both systems can be compared in a common codomain. This self-containedness is deliberate: importing inter-system relationships into $\mathcal{U}$ would require defining both systems categorically before constructing the category, reintroducing precisely the classificatory commitment CS is designed to avoid.

*Remark* 3.5 (Choice of state resolution). The choice of what counts as an observable state is a parameter of the model,-it determines the resolution at which $\mathcal{U}$ is constructed. CS is not invariant to this choice at the level of $\mathcal{U}$: different resolutions yield different categories. However, as established in Layer 3, CS is invariant at the level of the spectral signature: continuous changes in state resolution produce continuous changes in $\sigma(T)$, not discrete categorical jumps. The spectral signature is the invariant object; the state labelling is not.

### 3.2 Layer 2: The Representation Layer

Layer 2 maps the abstract morphic structure of $\mathcal{U}$ into a concrete operator-theoretic setting via a functor $F$. We first recall the necessary definitions from functional analysis.

**Definition 3.6** (Hilbert space and bounded operators)**.** A *Hilbert space* $\mathcal{H}$ is a complete inner product space over $\mathbb{R}$ or $\mathbb{C}$,-completeness ensures that every Cauchy sequence of vectors converges to a limit within $\mathcal{H}$. A *bounded linear operator* $T\colon \mathcal{H} \to \mathcal{H}$ is a linear map satisfying $\|Tv\| \leq C\,\|v\|$ for some constant $C > 0$ and all $v \in \mathcal{H}$. The *operator norm* of $T$ is $\|T\| = \sup_{\|v\|=1} \|Tv\|$, which is the maximum amplification $T$ can apply to any unit vector. The collection of all bounded linear operators on $\mathcal{H}$ is denoted $B(\mathcal{H})$; it is itself a Banach algebra under the operator norm. An operator $T \in B(\mathcal{H})$ is *self-adjoint* if $T = T^*$, where $T^*$ denotes the adjoint defined by $\langle Tv, w\rangle = \langle v, T^*w\rangle$ for all $v, w \in \mathcal{H}$.

**Definition 3.7** (The CS functor)**.** Let $\mathcal{U}$ be the category of undefined objects for a system $S$, and let **Hilb** denote the category of Hilbert spaces and bounded linear maps. The *CS functor* $F\colon \mathcal{U} \to$ **Hilb** is defined by:

(i) **On objects.** $F$ maps each abstract state $O \in \mathrm{Ob}(\mathcal{U})$ to a vector $F(O)$ in a complete Hilbert space $\mathcal{H}$, representing the state as a concrete mathematical object amenable to spectral analysis.

(ii) **On morphisms.** $F$ maps each morphism $f\colon O \to O'$ to a bounded linear operator $F(f) = T \in B(\mathcal{H})$, preserving the structure of the transition in the operator-theoretic setting.

(iii) **Functoriality,-identity preservation.** $F(\mathrm{id}_O) = \mathrm{id}_{F(O)}$ for all $O \in \mathrm{Ob}(\mathcal{U})$.

(iv) **Functoriality,-composition preservation.** $F(g \circ f) = F(g) \circ F(f)$ for all composable morphisms $f\colon O \to O'$ and $g\colon O' \to O''$.

*Remark* 3.8 (Status of functoriality). Conditions (iii) and (iv) of theorem 3.7 are the functoriality conditions. For the gravitational wave case,-the primary instantiation of CS in this paper,-$F$ maps spacetime configurations to the gravitational Hilbert space and morphisms to the linearised wave operator, and functoriality follows from the linearity of the perturbation equations (established in section 4). For general systems with inaccessible internal structure, the construction of a faithful functor $F$ satisfying the functoriality conditions is the primary open mathematical problem of the CS framework.

**Open Problem 3.9** (Functoriality in general)**.** Construct a functor $F\colon \mathcal{U} \to \mathbf{Hilb}$ satisfying theorem 3.7 for an arbitrary system with inaccessible internal structure, and prove that $F$ is *faithful*,-that is, that $F(f) = F(g)$ implies $f = g$ for all morphisms $f, g$ with the same domain and codomain. Faithfulness ensures that the functor does not collapse distinct transitions to the same operator, preserving the full morphic information of $\mathcal{U}$ in the operator-theoretic image.

**Definition 3.10** (Absolute boundary)**.** Let $T = F(f) \in B(\mathcal{H})$ be the operator image of a morphism $f$ under $F$. The *absolute boundary* of the system $S$ with respect to $f$ is the operator norm:

$$\|T\| = \sup_{\|v\|=1} \|Tv\| \,.$$

The absolute boundary is the physically grounded upper limit of the system's spectral extent. It is not imposed by the framework but derived from physical constraints on the system: for gravitational systems, $\|T\|$ is bounded by the speed of light $c$ and the Planck length $\ell_P$; for biological systems, $\|T\|$ is bounded by the metabolic ceiling of the system,-the maximum energy available to sustain state transitions. Boundedness of $T$ guarantees that $\|T\|$ is finite, which is essential for the spectral signature to have a well-defined upper limit.

*Remark* 3.11 (Self-adjointness assumption). The spectral analysis of Layer 3 requires $T$ to be self-adjoint or, more generally, normal,-satisfying $TT^* = T^*T$. Self-adjointness guarantees that $\sigma(T) \subset \mathbb{R}$ and that the spectral theorem applies in its standard form. CS assumes self-adjointness throughout; verifying this assumption for specific systems,-and extending the framework to non-self-adjoint operators via pseudospectral methods,-is an open problem.

### 3.3 Layer 3: The Spectral Layer

Layer 3 extracts the spectral signature of the system from the operator $T$ produced by Layer 2. We recall the relevant spectral theory before defining the signature.

**Definition 3.12** (Spectrum of a bounded operator)**.** Let $T \in B(\mathcal{H})$ be a bounded operator on a Hilbert space $\mathcal{H}$. The *spectrum* of $T$ is the set:

$$\sigma(T) = \{\lambda \in \mathbb{C} : (T - \lambda I) \text{ is not invertible in } B(\mathcal{H})\}.$$

For self-adjoint operators, $\sigma(T) \subset \mathbb{R}$. By the spectral theorem for bounded self-adjoint operators [Reed & Simon, 1972], $\sigma(T)$ is a non-empty compact subset of $\mathbb{R}$ contained within $[-\|T\|, \|T\|]$. For positive operators,-satisfying $\langle Tv, v\rangle \geq 0$ for all $v \in \mathcal{H}$,-$\sigma(T) \subset [0, \|T\|]$. The spectral measure $\mu_T$ associated to $T$ by the spectral theorem assigns to each Borel subset $E \subset \mathbb{R}$ a projection operator $\mu_T(E) \in B(\mathcal{H})$, and $T = \int \lambda \, d\mu_T(\lambda)$ in the sense of the Bochner integral. When $\mathcal{H}$ is infinite-dimensional, $\sigma(T)$ may be a continuous set,-not merely a discrete collection of eigenvalues,-admitting infinite internal density within $[0, \|T\|]$.

**Definition 3.13** (Spectral signature)**.** Let $S$ be a system with inaccessible internal structure, $\mathcal{U}$ its category of undefined objects, $F\colon \mathcal{U} \to \mathbf{Hilb}$ the CS functor, and $T = F(f) \in B(\mathcal{H})$ the operator image of a morphism $f$. The *spectral signature* of $S$ with respect to $f$ is the inner spectrum:

$$\sigma(T) \subset [0, \|T\|]$$

together with its associated spectral measure $\mu_T$. The spectral signature is the complete mathematical object CS assigns to $S$,-it replaces the categorical label as the fundamental descriptor of the system.

**Proposition 3.14** (Properties of the spectral signature)**.** *Let $T \in B(\mathcal{H})$ be a positive self-adjoint operator. The spectral signature $\sigma(T)$ satisfies:*

*(i)* ***Non-emptiness:*** *$\sigma(T) \neq \emptyset$.*

*(ii)* ***Compactness:*** *$\sigma(T)$ is a compact subset of $\mathbb{R}$.*

*(iii)* ***Boundedness:*** *$\sigma(T) \subset [0, \|T\|]$.*

*(iv)* ***Infinite internal density:*** *when $\mathcal{H}$ is infinite-dimensional, the spectral measure $\mu_T$ may assign positive measure to any open subinterval of $[0, \|T\|]$, admitting continuous gradations between any two spectral values.*

*Proof.* (i),(iii) follow directly from the spectral theorem for bounded self-adjoint operators [Reed & Simon, 1972, Theorem VII.1]. (iv) holds because in infinite-dimensional Hilbert space, the spectral theorem allows $T$ to have continuous spectrum,-a set of $\lambda$ for which

$(T - \lambda I)$ is not invertible but has dense range,-which assigns positive spectral measure to open intervals. □

*Remark* 3.15 (Finite vs. infinite-dimensional). For systems modelled on finite-dimensional Hilbert spaces,-finite graphs, finite state machines,-$\sigma(T)$ consists of finitely many eigenvalues and property (iv) does not hold in its strong form. CS applies to finite-dimensional cases but the claim of infinite internal density requires an infinite-dimensional embedding. The precise conditions under which a finite physical system admits an infinite-dimensional CS embedding is an open question.

**Proposition 3.16** (Unitary invariance of the spectral signature)**.** *Let $T \in B(\mathcal{H})$ be a positive self-adjoint operator and $U \in B(\mathcal{H})$ a unitary operator. Then $\sigma(UTU^*) = \sigma(T)$.*

*Proof.* $\lambda \in \sigma(T)$ if and only if $(T - \lambda I)$ is not invertible. Since $U$ is unitary, $UTU^* - \lambda I = U(T - \lambda I)U^*$, which is not invertible if and only if $(T - \lambda I)$ is not invertible, since $U$ and $U^*$ are themselves invertible. Hence $\sigma(UTU^*) = \sigma(T)$. □

### 3.4 The Spectral Complexity Index

The spectral signature $\sigma(T)$ is a distribution,-a rich mathematical object. To enable direct cross-domain comparison between systems whose functor images live in different Hilbert spaces, we define a scalar functional on $\sigma(T)$: the Spectral Complexity Index.

**Definition 3.17** (Quality factor)**.** Let mode $n$ of the spectral signature be characterised by oscillation frequency $f_n > 0$ and decay time $\tau_n > 0$. The *quality factor* of mode $n$ is:

$$Q_n = \pi f_n \tau_n.$$

$Q_n$ is a dimensionless scalar measuring the persistence of mode $n$: how many oscillation cycles occur before the mode decays by a factor of $e^{-\pi}$. A high $Q_n$ indicates a persistent, information-rich mode; a low $Q_n$ indicates a rapidly decaying, overdamped mode.

**Definition 3.18** (Spectral weights)**.** Let $\{A_n\}$ be the amplitudes of the spectral modes of $T$, where $A_n^2$ is proportional to the energy carried by mode $n$. The *spectral weight* of mode $n$ is:

$$p_n = \frac{A_n^2}{\sum_k A_k^2}.$$

The spectral weights satisfy $p_n \geq 0$ and $\sum_n p_n = 1$,-they form a probability distribution over the spectral modes.

**Definition 3.19** (Spectral Complexity Index)**.** Let $T \in B(\mathcal{H})$ be a positive self-adjoint operator with spectral modes $\{(f_n, \tau_n, A_n)\}$. The *Spectral Complexity Index* of $T$ is:

$$\mathrm{SCI}(T) = \sum_n p_n \cdot Q_n.$$

SCI($T$) is the amplitude-weighted mean quality factor across the spectrum,-the average persistence of the system's spectral modes, weighted by their energetic contribution. It is a scalar in $\mathbb{R}_{\geq 0}$.

**Proposition 3.20** (Non-negativity of SCI)**.** SCI($T$) $\geq 0$ *for all positive self-adjoint* $T \in B(\mathcal{H})$.

*Proof.* $p_n \geq 0$ by theorem 3.18 and $Q_n \geq 0$ by theorem 3.17, since $f_n > 0$ and $\tau_n > 0$ for all physical modes. Hence every term in the sum is non-negative. □

**Proposition 3.21** (Unitary invariance of SCI)**.** SCI($UTU^*$) = SCI($T$) *for all unitary* $U \in B(\mathcal{H})$.

*Proof.* By theorem 3.16, $\sigma(UTU^*) = \sigma(T)$. The spectral modes $\{(f_n, \tau_n, A_n)\}$ are determined entirely by $\sigma(T)$ and its associated spectral measure, both of which are invariant under unitary equivalence. Hence $p_n$ and $Q_n$ are invariant, and SCI($UTU^*$) $= \sum_n p_n \cdot Q_n =$ SCI($T$). □

theorem 3.21 is the key invariance result: SCI is independent of the choice of basis in $\mathcal{H}$. Two equivalent representations of the same physical system,-related by a unitary transformation,-produce identical SCI values. This ensures that SCI measures a property of the system, not of the representation.

**Open Problem 3.22** (Monotonicity of SCI)**.** Define a partial order on spectral distributions by $\mu \leq \mu'$ if $\mu'$ is obtained from $\mu$ by redistributing spectral weight toward higher-$Q$ modes. We conjecture that SCI is monotone with respect to this order,-that richer spectral distributions yield higher SCI values. A formal proof requires a precise definition of the partial order and a monotonicity argument for the weighted sum.

**Open Problem 3.23** (Boundedness of SCI)**.** We conjecture that SCI($T$) $\leq \|T\| \cdot \max_n Q_n$ for all positive self-adjoint $T \in B(\mathcal{H})$. An upper bound on SCI in terms of the absolute boundary $\|T\|$ would ensure that SCI is always finite for bounded operators and would provide a natural normalisation.

*Remark* 3.24 (Relationship to spectral entropy)*.* The spectral entropy of $T$ is defined as $H(T) = -\sum_n p_n \log p_n$. Spectral entropy measures the distributional spread of spectral weight but discards temporal information,-it assigns equal significance to modes regardless of their persistence. SCI incorporates $Q_n$ precisely to capture persistence. Spectral entropy is a limiting case of SCI: when all $Q_n$ are equal to a constant $Q$, SCI $= Q \cdot \sum_n p_n = Q$, a constant independent of the distribution. For systems with heterogeneous $Q_n$,-which is the generic case for physical systems,-SCI is strictly more informative than spectral entropy.

*Remark* 3.25 (Cross-domain comparability)*.* SCI takes values in $\mathbb{R}_{\geq 0}$ regardless of which Hilbert space $\mathcal{H}$ the functor $F$ targets. Two systems $S$ and $S'$ whose functors land in different Hilbert spaces,-$L^2$(cortical surface) for a biological system, a gravitational Hilbert

space for a black hole,-nonetheless have SCIs that are directly comparable as real numbers. The cross-domain comparison that is the central claim of CS occurs in $\mathbb{R}$, not in **Hilb**. The Hilbert spaces are intermediate machinery; $\mathbb{R}$ is the comparison space.

# 4 Instantiation: Gravitational Wave Identification

This section applies the CS framework to the identification of gravitational wave sources. We construct $\mathcal{U}$, the functor $F$, the spectral signature, and the SCI for the binary black hole merger event GW150914 [Abbott et al., 2016]. All numerical values are derived from published LIGO parameter estimates, and the SCI computation is exact within the precision of those estimates. No new empirical data is introduced.

## 4.1 Simple Instantiation: The Resonant Black Box

Before applying the CS framework to the gravitational case, we illustrate the three-layer procedure on a deliberately minimal system. The purpose of this example is pedagogical: to show that the framework operates on observable morphic structure alone, without requiring knowledge of internal composition.

**The system.** Consider a physical system $S$,-a sealed, opaque enclosure of unknown internal composition,-that, when excited, produces a damped oscillatory output observable at its boundary. The internal structure of $S$ is inaccessible in the sense of theorem 3.1: we observe only the time-domain output signal $h(t)$ at the boundary. We do not know whether $S$ contains a spring-mass system, an electrical LC circuit, an acoustic resonator, or something else entirely. CS does not require us to know.

**Layer 1.** We construct $\mathcal{U}$ for $S$. *Objects*: two abstract states,-$O$ (excited state, producing oscillatory output) and $O_0$ (quiescent state, no output). *Morphism*: one non-trivial morphism $f\colon O \to O_0$,-the transition observable as the decay of $h(t)$ from its peak amplitude to zero. By theorem 3.3, $\mathcal{U}$ is a well-defined small category.

**Layer 2.** We apply $F\colon \mathcal{U} \to$ **Hilb**. $F$ maps both $O$ and $O_0$ to $L^2(\mathbb{R})$.[1] $F$ maps $f\colon O \to O_0$ to the bounded linear operator $T \in B(L^2(\mathbb{R}))$ defined by:

$$(Th)(t) = e^{-t/\tau} \cos(2\pi f_1 t) \cdot h(0),$$

where $f_1$ is the observed oscillation frequency and $\tau_1$ is the observed decay time, both read directly from $h(t)$. $T$ is bounded with $\|T\| = 1$.

[1] $F$ assigns the same Hilbert space $L^2(\mathbb{R})$ to both objects. Faithfulness holds because $F(f) = T \neq \mathrm{id}_{L^2(\mathbb{R})} = F(\mathrm{id}_O)$, so $F$ distinguishes the non-trivial morphism from the identity.

**Layer 3 and SCI.** The operator $T$ has a single spectral mode with spectral weight $p_1 = 1$, quality factor $Q_1 = \pi f_1 \tau_1$, and:

$$\mathrm{SCI}(S) = p_1 \cdot Q_1 = \pi f_1 \tau_1.$$

**Numerical example.** For $f_1 = 100\,\mathrm{Hz}$, $\tau_1 = 5\,\mathrm{ms}$: $\mathrm{SCI}(S) = \pi \times 100 \times 0.005 = 1.57$. For a second enclosure $S'$ with $f_1 = 200\,\mathrm{Hz}$, $\tau_1 = 5\,\mathrm{ms}$:

$$\mathrm{SCI}(S') = \pi \times 200 \times 0.005 = 3.14.$$

We note that $\mathrm{SCI}(S') = 3.14$ coincides with $\mathrm{SCI}(\mathrm{GW150914}) = 3.14$, computed in section 4.5 from the quasinormal mode spectrum of a merging black hole system. This coincidence is not engineered,-it arises from the parameters chosen for this pedagogical example independently of the gravitational computation. Its significance is precisely what Conjecture 6.4 addresses: two systems with inaccessible internal structure, drawn from maximally dissimilar physical domains, placed at the same point in spectral complexity space by the same formula.

*Remark* 4.1. The resonant black box satisfies theorem 3.1 by construction. The same procedure applies to any system from which a damped oscillatory output can be observed,-regardless of whether the system is physically sealed. A brain observed only through its EEG output, a star observed only through its asteroseismic oscillations, an economy observed only through its price oscillations,-all satisfy the morphic observability condition required by $\mathcal{U}$ and all admit SCI computation from their output spectra. The black box framing is pedagogical; the scope is general. Among these cases, gravity is chosen as the primary instantiation because it provides an exact computation from published LIGO data, whereas the others currently require estimation from empirical proxies.

### 4.2 Constructing $\mathcal{U}$ for the Gravitational System

**Definition 4.2** (Gravitational category $\mathcal{U}_{\mathrm{grav}}$)**.** Let $S$ be the spacetime geometry in the neighbourhood of a binary black hole merger. The category $\mathcal{U}_{\mathrm{grav}}$ is constructed as follows:

(i) **Objects.** Configurations of the spacetime metric $g_{\mu\nu}$,-observationally distinguishable geometric states characterised by the metric perturbation $h_{\mu\nu}$ at spatial infinity. No definition of what a black hole fundamentally is, no specification of the interior, is required.

(ii) **Morphisms.** Causal dynamical transitions between metric configurations,-geodesic deviation, gravitational wave emission, and the irreversible transition from two-body inspiral to single-body ringdown. These transitions are observable at spatial infinity via the gravitational wave strain $h(t)$ measured by LIGO detectors.

(iii) **Identity, composition, associativity.** Verified by theorem 3.3.

*Remark* 4.3 (Inaccessibility in the gravitational case). The inaccessibility of theorem 3.1 is satisfied maximally here. By Penrose's singularity theorem [Penrose, 1965] and the causal structure of Kerr spacetime, no information from the interior of either black hole reaches spatial infinity. $\mathcal{U}_{\text{grav}}$ is constructed without any assumption about the interior.

### 4.3 The Functor $F$ for the Gravitational System

**Definition 4.4** (Gravitational Hilbert space)**.** Let $\mathcal{H}_{\text{grav}} = L^2(\mathbb{R}^3, g)$ be the Hilbert space of square-integrable perturbations of the background Kerr metric, with inner product defined by the background metric $g$. Metric perturbations $h_{\mu\nu}$ satisfying appropriate gauge conditions and boundary conditions form a dense subspace of $\mathcal{H}_{\text{grav}}$.

**Definition 4.5** (Gravitational functor $F_{\text{grav}}$)**.** The functor $F_{\text{grav}} \colon \mathcal{U}_{\text{grav}} \to \mathbf{Hilb}$ is defined by:

(i) **On objects.** Each metric configuration state $O \in \text{Ob}(\mathcal{U}_{\text{grav}})$ maps to the corresponding perturbation vector $h_{\mu\nu} \in \mathcal{H}_{\text{grav}}$.

(ii) **On morphisms.** Each causal transition $f \colon O \to O'$ maps to the linearised Einstein field operator $\Box h_{\mu\nu} = -16\pi G\, T_{\mu\nu}$, where $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ is the covariant wave operator acting on $\mathcal{H}_{\text{grav}}$.

**Proposition 4.6** (Functoriality of $F_{\text{grav}}$)**.** *$F_{\text{grav}}$ satisfies the functoriality conditions of theorem 3.7.*

*Proof sketch.* Identity preservation holds because the trivial transition,-no change in metric configuration,-maps to the identity operator on $\mathcal{H}_{\text{grav}}$. Composition preservation holds because the linearised Einstein equations are linear: the composition of two causal transitions maps to the composition of the corresponding wave operators, which is a consequence of the linearity of perturbation theory. Boundedness of $\Box$ as an operator on $\mathcal{H}_{\text{grav}}$ holds under appropriate boundary conditions at spatial infinity, ensuring $\Box \in B(\mathcal{H}_{\text{grav}})$. A full proof of boundedness requires specification of the functional analytic domain of $\Box$, which we take as established in the linearised gravity literature [Wald, 1984]. $\Box$

*Remark* 4.7 (Absolute boundary in the gravitational case). The operator norm $\|\Box\|$ is bounded by the physical constraints of the system: the speed of light $c$ sets the maximum propagation speed of metric perturbations, and the Planck length $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}\,\text{m}$ sets the minimum length scale below which classical general relativity is no longer valid. $\|\Box\|$ is finite within this regime.

### 4.4 The Spectral Signature: Quasinormal Modes

**Definition 4.8** (Quasinormal modes)**.** A *quasinormal mode* of a Kerr black hole with mass $M$ and dimensionless spin $\chi = a/M$ is a solution to the linearised perturbation equations of the form:

$$h_{\mu\nu}(t) = \mathrm{Re}\Big[A_n e^{-i\omega_n t}\Big]\,,$$

where $\omega_n = 2\pi f_n - i/\tau_n$ is a complex frequency with real part $2\pi f_n$ giving the oscillation frequency and imaginary part $1/\tau_n$ giving the inverse decay time. The quasinormal mode spectrum $\{\omega_n\}$ constitutes $\sigma(\Box)$. Quasinormal modes are labelled by three integers $(l, m, n)$ corresponding to angular momentum, azimuthal, and overtone quantum numbers respectively.

*Remark* 4.9 (Self-adjointness in the gravitational case). Quasinormal modes are not eigenfunctions of a self-adjoint operator in the standard $L^2$ sense,-they satisfy outgoing boundary conditions at spatial infinity and ingoing conditions at the horizon, which makes the boundary value problem non-self-adjoint. The quasinormal frequencies $\omega_n$ are therefore complex rather than real. We handle this as follows: the physically relevant quantities are the real part $f_n$ and the imaginary part $1/\tau_n$ of $\omega_n$, both of which are real and positive. The quality factor $Q_n = \pi f_n \tau_n$ is constructed directly from these real quantities, bypassing the non-self-adjointness issue. A rigorous treatment would require extending CS to pseudospectral methods for non-self-adjoint operators,-an open problem noted in theorem 3.11,-but for the purposes of SCI computation the quality factor construction is well-defined.

### 4.5 SCI Computation for GW150914

We apply theorems 3.17 to 3.19 to compute SCI for GW150914.

The final black hole parameters are taken from the LIGO parameter estimation of Abbott et al. [2016]: final mass $M_f = 68.5\,M_\odot$, dimensionless spin $\chi_f = 0.67$, with uncertainties of approximately $\pm 4\,M_\odot$ and $\pm 0.05$ respectively at 90% credible interval.

The dominant quasinormal mode $(l = 2, m = 2, n = 0)$ from Kerr perturbation theory [Berti et al., 2006] for $M_f = 68.5\,M_\odot$ and $\chi_f = 0.67$: oscillation frequency $f_{220} = 251\,\mathrm{Hz}$, decay time $\tau_{220} = 4.0\,\mathrm{ms}$, quality factor $Q_{220} = \pi \times 251 \times 0.004 = 3.16$.

Spectral weights from ringdown energy analysis: $p_{220} \approx 0.97$, $p_{330} + p_{221} \approx 0.03$. Higher mode quality factors [Berti et al., 2006]: $Q_{330} \approx 2.80$, $Q_{221} \approx 1.62$.

$$
\begin{aligned}
\mathrm{SCI}(\mathrm{GW150914}) &= p_{220} \cdot Q_{220} + p_{330} \cdot Q_{330} + p_{221} \cdot Q_{221} \\
&\approx 0.97 \times 3.16 + 0.02 \times 2.80 + 0.01 \times 1.62 \\
&\approx 3.065 + 0.056 + 0.016 \\
&\approx 3.14.
\end{aligned}
$$

**Proposition 4.10** (SCI of GW150914)**.** *Under the CS framework, with the gravitational functor $F_{\mathrm{grav}}$ of theorem 4.5, the Spectral Complexity Index of the GW150914 post-merger system is:*

$$\mathrm{SCI}(\mathrm{GW150914}) = 3.14 \pm 0.08,$$

*where the uncertainty propagates from the 90% credible interval uncertainties in $M_f$ and $\chi_f$ reported by Abbott et al. [2016].*

*Proof.* Direct computation from theorems 3.17 to 3.19 using the quasinormal mode parameters of theorem 4.8 with values from Berti et al. [2006] and LIGO parameter estimates from Abbott et al. [2016].[2] □

### 4.6 Interpretation and Predictions

SCI(GW150914) = 3.14 places the post-merger black hole at a specific point in spectral complexity space. Three observations follow.

SCI varies continuously and predictably with the spin parameter $\chi$. For fixed mass $M_f$, increasing $\chi$ toward the extremal Kerr limit $\chi \to 1$ increases $f_{220}$ and $\tau_{220}$,-hence $Q_{220}$ and SCI both increase monotonically. This constitutes a precise falsifiable prediction: across the LIGO catalogue of binary black hole mergers, SCI should increase monotonically with the dimensionless spin of the final black hole. Verification of this prediction using published LIGO parameter estimates is deferred to subsequent work.

Second, the dominant mode approximation $p_{220} \approx 0.97$ reflects the physical fact that binary black hole mergers with near-equal masses and low precession excite the fundamental mode almost exclusively. Highly asymmetric mass ratios or precessing systems excite higher harmonics more strongly, increasing SCI for a given total mass and spin. This constitutes

[2]The uncertainty $\pm 0.08$ is propagated as follows. The dominant mode quality factor $Q_{220} = \pi f_{220} \tau_{220}$ depends on $M_f$ and $\chi_f$ through the dimensionless quasinormal mode frequency $M\omega_{220}$ tabulated in Berti et al. [2006]. At $M_f = 68.5 \pm 4\, M_\odot$ and $\chi_f = 0.67 \pm 0.05$, linear error propagation gives $\delta Q_{220} \approx 0.07$ from the spin uncertainty,-spin is the dominant source of uncertainty since $f_{220}$ and $\tau_{220}$ both increase with $\chi$. The mass uncertainty contributes negligibly to $Q_{220}$ since $M\omega_{220}$ is dimensionless and $M_f$ enters only through the physical frequency $f_{220} = M\omega_{220}/(2\pi G M_f/c^3)$, which cancels in the product $f_{220} \cdot \tau_{220}$. The spectral weight $p_{220} \approx 0.97$ is taken as fixed since subdominant mode contributions are below the measurement precision of current LIGO data. Hence $\delta\mathrm{SCI} \approx p_{220} \cdot \delta Q_{220} \approx 0.97 \times 0.07 \approx 0.07$, rounded to $\pm 0.08$.

a second prediction: higher mass-ratio mergers should exhibit higher SCI for equivalent final state parameters.

Third, SCI(GW150914) = 3.14 is computed without any categorical commitment to the nature of the source. If the no-hair theorem holds,-if the quasinormal spectrum is uniquely determined by mass and spin,-then SCI is uniquely determined by $(M_f, \chi_f)$. If the no-hair theorem fails,-if the spectrum contains additional modes from exotic compact objects,-SCI will differ from the Kerr prediction, providing a spectroscopic test of the no-hair theorem via CS.

## 5 Proposed Application: Consciousness

This section applies the CS framework to consciousness science. Unlike section 4, no new empirical computation is performed. The application is explicitly theoretical: we construct $\mathcal{U}$, propose $F$, define the spectral signature, and advance SCI as a theoretically grounded measure of conscious complexity. Empirical validation is deferred to subsequent work. We are explicit about this throughout.

### 5.1 Why Consciousness Satisfies theorem 3.1

Consciousness satisfies the inaccessibility condition in a stronger sense than most physical systems. First, the internal states of a conscious system,-phenomenal states, qualia, the subjective character of experience,-are not directly observable by any external instrument. Second, the definitional question,-what consciousness is,-has resisted resolution for centuries despite sustained philosophical and scientific attention [Chalmers, 1995, Nagel, 1974]. Third, Tsuchiya and Saigo [Tsuchiya et al., 2016, Tsuchiya & Saigo, 2021] argued from within consciousness science that the appropriate mathematical framework for consciousness is categorical and functorial,-that phenomenal structures form a category and that their relationships require a functor to a grounding mathematical space. CS provides that functor.

### 5.2 Constructing $\mathcal{U}$ for Consciousness

**Definition 5.1** (Consciousness category $\mathcal{U}_{\text{con}}$)**.** Let $S$ be a conscious system. The category $\mathcal{U}_{\text{con}}$ is constructed as follows:

(i) **Objects.** Phenomenal states of $S$,-observationally distinguishable configurations of conscious experience, including wakefulness, deep sleep, anaesthesia, dreaming, and pharmacologically altered states. The objects of $\mathcal{U}_{\text{con}}$ are not defined by what consciousness is,-they are identified by their observational distinguishability established through behavioural and neural correlate evidence.

(ii) **Morphisms.** Observable transitions between phenomenal states,-including the transition from wakefulness to sleep, from consciousness to anaesthesia, from baseline to pharmacologically altered states. Each transition is observable through behavioural markers and neural correlate measurements without requiring knowledge of the phenomenal character of either state.

(iii) **Identity, composition, associativity.** Verified by theorem 3.3.

*Remark* 5.2. theorem 5.1 makes no claim about what consciousness is. The objects of $\mathcal{U}_{\text{con}}$ are identified by observational distinguishability, not by any prior theory of phenomenal character. The definition, if it exists, emerges from the spectral structure rather than being imposed in advance.

### 5.3 The Functor $F$ for Consciousness

**Definition 5.3** (Proposed consciousness functor $F_{\text{con}}$)**.** Let $\mathcal{H}_{\text{con}} = L^2(\Omega)$ be the Hilbert space of square-integrable functions on the cortical surface $\Omega$. The proposed functor $F_{\text{con}} \colon \mathcal{U}_{\text{con}} \to \mathbf{Hilb}$ is defined by:

(i) **On objects.** Each phenomenal state $O \in \text{Ob}(\mathcal{U}_{\text{con}})$ maps to the corresponding neural activation pattern $\psi_O \in L^2(\Omega)$,-the spatial distribution of neural activity across the cortical surface in state $O$, measurable via fMRI or EEG source reconstruction.

(ii) **On morphisms.** Each transition $f \colon O \to O'$ maps to the graph Laplacian operator $L = D - A \in B(L^2(\Omega))$, where $A$ is the structural connectivity matrix of the cortical connectome and $D$ is the diagonal degree matrix.

**Open Problem 5.4** (Functoriality of $F_{\text{con}}$)**.** Verify that $F_{\text{con}}$ satisfies the functoriality conditions of theorem 3.7: that $F_{\text{con}}(\text{id}_O) = \text{id}_{L^2(\Omega)}$ and that $F_{\text{con}}(g \circ f) = F_{\text{con}}(g) \circ F_{\text{con}}(f)$ for all composable morphisms in $\mathcal{U}_{\text{con}}$.

*Remark* 5.5 (Self-adjointness in the consciousness case). The graph Laplacian $L = D - A$ is symmetric by construction when $A$ is symmetric,-that is, when the connectome is modelled as an undirected graph with symmetric connection weights. In this case $L$ is self-adjoint on $L^2(\Omega)$ and the spectral theorem applies directly. For directed connectome models,-which better capture the asymmetric effective connectivity of the brain,-$L$ is no longer symmetric and self-adjointness fails. We note that the undirected approximation is standard in the connectome harmonics literature [Atasoy et al., 2016] and that the directed extension is an open problem shared by all spectral connectome methods,-it is not a gap specific to CS but a limitation of the current empirical framework that CS inherits rather than introduces.

### 5.4 The Spectral Signature: Connectome Harmonics

**Definition 5.6** (Connectome harmonics)**.** Let $L = D - A$ be the graph Laplacian of the cortical connectome with eigendecomposition $L\varphi_n = \lambda_n \varphi_n$, where $0 = \lambda_0 \leq \lambda_1 \leq \cdots \leq \lambda_N$

are the eigenvalues and $\{\varphi_n\}$ are the corresponding eigenfunctions on the cortical surface. The connectome harmonics $\{\varphi_n\}$ are the natural modes of neural activity propagation on the structural connectome,-they play the same role for the brain that Fourier modes play for a homogeneous medium, but adapted to the irregular geometry and connectivity of the cortical surface [Atasoy et al., 2016].

*Remark* 5.7 (Empirical basis). Connectome harmonics were introduced and empirically validated by Atasoy et al. [2016], who demonstrated that resting-state fMRI activity decomposes into a small number of dominant harmonics and that different consciousness states,-wakefulness, deep sleep, LSD, anaesthesia,-produce characteristically different harmonic energy distributions. This empirical result constitutes independent validation of the Layer 3 structure of CS for the consciousness case.

### 5.5 Proposed SCI for Consciousness

**Definition 5.8** (Consciousness SCI)**.** Let $\{f_n^{\mathrm{BOLD}}, \tau_n, E_n\}$ be the connectome harmonic parameters for conscious system $S$ in state $O$. The spectral weights are $p_n = E_n / \sum_k E_k$, the quality factor of harmonic $n$ is $Q_n = \pi f_n^{\mathrm{BOLD}} \tau_n$, and the proposed SCI for $S$ in state $O$ is:

$$\mathrm{SCI}_{\mathrm{con}}(O) = \sum_n p_n \cdot Q_n.$$

### 5.6 Theoretical Advantages of SCI over Existing Measures

We advance three theoretical arguments for why SCI is a superior measure of conscious complexity relative to existing frameworks, on structural grounds independent of empirical validation.

**Superiority over $\Phi$.** Integrated Information Theory's $\Phi$ [Tononi, 2004, Oizumi et al., 2014] requires full enumeration of the system's elements, states, and causal relationships,-a combinatorial computation that is intractable for systems beyond a small number of elements. SCI requires only the spectral decomposition of the connectome Laplacian, computable in polynomial time via standard eigendecomposition algorithms. SCI is therefore both computationally tractable and applicable where $\Phi$ is not.

**Superiority over PCI.** The Perturbational Complexity Index [Casali et al., 2013] is an empirically derived measure that lacks a theoretical foundation in a formal mathematical framework. SCI is derived directly from the functorial structure of CS: it is not a heuristic but a formal consequence of the spectral analysis of the functor image. SCI therefore provides what PCI lacks: a theoretical grounding that connects the empirical measure to a mathematical framework with precise axioms and a stated central conjecture.

**Prediction 5.9** (SCI ordering matches PCI ordering)**.** For any pair of conscious states $O$ and $O'$ for which $\mathrm{PCI}(O) < \mathrm{PCI}(O')$, CS predicts $\mathrm{SCI}_{\mathrm{con}}(O) < \mathrm{SCI}_{\mathrm{con}}(O')$. This prediction is falsifiable by computing $\mathrm{SCI}_{\mathrm{con}}$ from published connectome harmonic data for the consciousness states studied in Casali et al. [2013] and Atasoy et al. [2016]. Verification or falsification of this prediction is the primary proposed empirical test of the consciousness application of CS, deferred to subsequent work.

**Substrate independence.** Both $\Phi$ and PCI are defined for specific physical substrates. SCI is substrate-independent by construction,-theorem 3.21 establishes that SCI is invariant under unitary equivalence, and theorem 3.25 establishes that SCI values from different Hilbert spaces are directly comparable as real numbers. This is a theoretical advantage that neither $\Phi$ nor PCI possesses and that follows necessarily from the CS framework rather than being added as an assumption.

### 5.7 Open Problems for the Consciousness Application

**Open Problem 5.10** (Directed connectome extension)**.** Extend CS to directed connectome models using non-self-adjoint spectral theory or pseudospectral methods.

**Open Problem 5.11** (SCI,$\Phi$ relationship)**.** Determine whether $\mathrm{SCI} = f(\Phi)$ for some function $f$ when the conscious system is fully enumerable.

*Remark* 5.12 (The hard problem and CS)*.* The hard problem of consciousness [Chalmers, 1995] asks why physical processes give rise to subjective experience. CS does not solve the hard problem. What CS does is reframe it mathematically: rather than asking "why does this category of physical process give rise to experience?", the question becomes "what spectral complexity threshold is necessary and sufficient for phenomenal character to arise?" This is a sharper and potentially more tractable formulation,-it transforms a conceptual puzzle into a mathematical problem with a defined proof target. We do not claim the spectral formulation resolves the hard problem. We claim it makes the problem mathematically precise in a way that categorical formulations do not.

## 6 The Central Conjecture

This section states the central mathematical claim of Categorical Spectralism. The conjecture is stated with full formal precision, its proof target is enumerated as a structured list of required lemmas, and its falsification conditions are made explicit.

### 6.1 Preliminary Definitions

**Definition 6.1** (Categorical equivalence via functor images)**.** Let $A$ and $B$ be systems with inaccessible internal structure, with functor images $F(A) = T_A \in B(\mathcal{H}_A)$ and

$F(B) = T_B \in B(\mathcal{H}_B)$ respectively. $A$ and $B$ are *categorically equivalent* in the CS sense if there exists a unitary operator $V\colon \mathcal{H}_A \to \mathcal{H}_B$ such that:

$$VT_AV^* = T_B.$$

That is, $T_A$ and $T_B$ are unitarily equivalent.

*Remark* 6.2 (Why unitary equivalence). Unitary equivalence is chosen as the definition of categorical equivalence for three reasons. First, it is the standard notion of isomorphism in operator theory. Second, by theorem 3.21, SCI is invariant under unitary equivalence, making the conjecture well-posed. Third, unitary equivalence is strictly stronger than spectral equivalence,-two operators may have identical spectra without being unitarily equivalent,-which means the conjecture, if true, asserts that SCI captures information beyond the spectrum alone, or alternatively that the spectrum determines the unitary equivalence class for the class of operators arising in CS.

**Definition 6.3** (SCI equivalence)**.** Two systems $A$ and $B$ are *SCI-equivalent* if $\mathrm{SCI}(T_A) = \mathrm{SCI}(T_B)$ as real numbers.

### 6.2 The Central Conjecture

**Conjecture 6.4** (Categorical equivalence via spectral complexity)**.** *Let $A$ and $B$ be systems with inaccessible internal structure admitting the CS functor construction of theorem 3.7, with operator images $T_A \in B(\mathcal{H}_A)$ and $T_B \in B(\mathcal{H}_B)$ respectively. If $\mathrm{SCI}(T_A) = \mathrm{SCI}(T_B)$, then $T_A$ and $T_B$ are unitarily equivalent,-that is, $A$ and $B$ are categorically equivalent in the sense of theorem 6.1. Formally:*

$$\mathrm{SCI}(T_A) = \mathrm{SCI}(T_B) \implies \exists \text{ unitary } V\colon \mathcal{H}_A \to \mathcal{H}_B \text{ such that } VT_AV^* = T_B.$$

*Remark* 6.5 (Directionality). theorem 6.4 is stated in one direction only: SCI equivalence implies categorical equivalence. The converse,-categorical equivalence implies SCI equivalence,-follows immediately from theorem 3.21: if $T_A$ and $T_B$ are unitarily equivalent then $\sigma(T_A) = \sigma(T_B)$ and hence $\mathrm{SCI}(T_A) = \mathrm{SCI}(T_B)$. The converse is therefore already proven. theorem 6.4 asserts the non-trivial direction: that SCI equivalence is sufficient for categorical equivalence, not merely necessary.

*Remark* 6.6 (Strength of the claim). theorem 6.4 is a strong claim. It asserts that a single real number,-SCI,-determines the unitary equivalence class of the operator $T$, up to the class of operators arising from the CS functor construction. For general operators in $B(\mathcal{H})$, this is false: two operators may have identical SCI without being unitarily equivalent. The conjecture therefore implicitly restricts to the class of CS operators,-operators arising as functor images of morphic structure,-and asserts that within this class, SCI is a complete invariant. Characterising this class precisely is part of the proof programme.

### 6.3 What a Proof Requires

A proof of theorem 6.4 requires establishing the following lemmas in sequence.

**Lemma 6.7** (CS operators form a restricted class)**.** *Characterise the class $\mathcal{C}_{\mathrm{CS}} \subset B(\mathcal{H})$ of operators that arise as functor images $F(f)$ for some morphism $f$ in some category $\mathcal{U}$ of undefined objects. Show that $\mathcal{C}_{\mathrm{CS}}$ is closed under the operations relevant to the conjecture,-in particular, that unitary equivalence preserves membership in $\mathcal{C}_{\mathrm{CS}}$.*

Status: open. *This is the foundational characterisation problem. Without knowing what class of operators CS produces, the conjecture cannot be stated with full precision.*

**Lemma 6.8** (SCI determines the spectrum for CS operators)**.** *Show that for operators $T \in \mathcal{C}_{\mathrm{CS}}$, $\mathrm{SCI}(T) = \mathrm{SCI}(T')$ implies $\sigma(T) = \sigma(T')$ as spectral measures,-not merely as sets of eigenvalues but as full spectral distributions including multiplicities and spectral measures.*

Status: open. *This requires showing that the SCI functional is injective on the space of spectral measures of CS operators.*

**Lemma 6.9** (Spectral equivalence implies unitary equivalence for CS operators)**.** *Show that for operators $T, T' \in \mathcal{C}_{\mathrm{CS}}$, $\sigma(T) = \sigma(T')$ as spectral measures implies $T$ and $T'$ are unitarily equivalent.*

Status: partially known. *For self-adjoint operators on separable Hilbert spaces, the spectral theorem provides a complete classification by spectral measure,-two self-adjoint operators with the same spectral measure are unitarily equivalent [Reed & Simon, 1972, Theorem VII.6]. If CS operators are self-adjoint and their Hilbert spaces are separable,-both conditions assumed in section 4,-this lemma follows from existing results.*

**Lemma 6.10** (SCI is injective on CS spectral measures)**.** *Show that the map* SCI: {*spectral measures of C* $\mathbb{R}_{\geq 0}$ *is injective,-that is, that* $\mathrm{SCI}(T) = \mathrm{SCI}(T')$ *implies the spectral measures of $T$ and $T'$ are identical.*

Status: open. *This is the most technically demanding component of the proof. Injectivity of SCI on spectral measures requires showing that the amplitude-weighted mean quality factor functional uniquely determines the full spectral distribution within the class of CS operators.*

*Remark* 6.11 (Proof strategy). The natural proof strategy combines theorems 6.7 to 6.10 as follows:

(i) $\mathrm{SCI}(T_A) = \mathrm{SCI}(T_B)$

(ii) $\to \sigma(T_A) = \sigma(T_B)$ as spectral measures (theorem 6.10)

(iii) $\to T_A$ and $T_B$ are unitarily equivalent (theorem 6.9)

(iv) $\to A$ and $B$ are categorically equivalent (theorem 6.1)

Step (i)→(ii) via theorem 6.10 is the primary open problem. Step (ii)→(iii) requires verifying the self-adjointness and separability conditions of theorem 6.9 for CS operators, which connects to theorem 3.9.

### 6.4 Falsification Conditions

**Definition 6.12** (Mathematical falsification)**.** theorem 6.4 is falsified if there exist two systems $A$ and $B$ with $\mathrm{SCI}(T_A) = \mathrm{SCI}(T_B)$ but $T_A$ and $T_B$ are not unitarily equivalent. The most natural candidate for a counterexample is a pair of operators with the same SCI but different spectral distributions. Whether such operators arise within the class $\mathcal{C}_{\mathrm{CS}}$ is the question theorem 6.10 addresses.

**Definition 6.13** (Empirical falsification)**.** theorem 6.4 is empirically falsified if two systems with equal measured SCI are shown to be categorically distinguishable by some observable property. The most direct empirical test is: compute SCI for pairs of conscious states with known categorical relationships and determine whether SCI equivalence tracks categorical equivalence.

**Definition 6.14** (Domain restriction falsification)**.** A weaker falsification is possible: theorem 6.4 holds within a single domain,-gravity or consciousness,-but fails across domains. The coincidence SCI(GW150914) $\approx$ SCI(black box) $\approx 3.14$ noted in section 4 constitutes the first cross-domain test case.

### 6.5 Implications if True

If theorem 6.4 is true, SCI is a complete invariant of the categorical equivalence class for systems with inaccessible internal structure. This has three consequences.

First, categorical classification of systems with inaccessible internal structure is formally redundant wherever SCI is defined. Classification persists as a practical tool but loses its status as a fundamental mathematical object.

Second, substrate independence of conscious complexity follows as a theorem rather than an assumption. Two systems,-biological and artificial, carbon and silicon,-with equal SCI are categorically equivalent in the CS sense. The substrate independence of consciousness, currently a contested philosophical claim, becomes a mathematical consequence of the conjecture.

Third, the cross-domain comparability established by theorem 3.25 acquires full mathematical force. The coincidence SCI(GW150914) $\approx$ SCI(black box) $\approx 3.14$ would not merely be a numerical curiosity but a statement of categorical equivalence between a merging black hole system and any other system at the same spectral complexity,-a striking and counterintuitive unification of physical domains.

### 6.6 Relationship to the Open Problems of section 4

theorem 6.4 is not independent of the open problems stated in section 4,-it depends on their resolution. theorem 3.9 (functoriality in general) is a prerequisite for theorem 6.7. theorem 3.22 (monotonicity of SCI) would strengthen theorem 6.10. theorem 3.23 (boundedness of SCI) ensures that the map $\mathrm{SCI}\colon \mathcal{C}_{\mathrm{CS}} \to \mathbb{R}_{\geq 0}$ has compact range. The proof programme of this section therefore depends on prior resolution of theorems 3.9, 3.22 and 3.23.

## 7 Discussion

### 7.1 CS as Mathematical Structuralism

CS is a mathematical implementation of ontic structural realism,-the philosophical position, developed by Ladyman [Ladyman, 1998] and Worrall [Worrall, 1989], that what exists fundamentally are structures and relations rather than intrinsically characterised objects. On this view, the electron is not a thing with intrinsic properties,-it is a node in a web of structural relationships encoded in the mathematics of quantum field theory. The particle label is a derivative notion, not a primitive one.

CS formalises this position for systems with inaccessible internal structure. The objects of $\mathcal{U}$ are defined entirely by their morphic relationships,-their identity is exhausted by the transitions they participate in. The spectral signature $\sigma(T)$ is the mathematical object that ontic structural realism has always needed but never had: a substrate-independent, continuously valued invariant that encodes structural relationships without presupposing an underlying substrate. If ontic structural realism is correct, then theorem 6.4 is not merely plausible but expected. The mathematical work required is to make the structural equivalence precise enough to prove.

### 7.2 CS and the Philosophy of Scientific Measurement

A persistent objection to continuous complexity measures,-of which SCI is one,-is the measurement problem: what does it mean to say that two systems at the same SCI are equivalent, when the SCI is computed from different physical quantities in each case? For the black hole, SCI is computed from gravitational wave frequencies and decay times. For the resonant black box, SCI is computed from acoustic oscillation parameters. The numbers happen to coincide. But do they mean the same thing?

CS has a precise answer. SCI is not a measurement of a physical quantity in the usual sense,-it is a functional on the spectral measure of an operator, and the operator is the functor image of the system's morphic structure. Two systems at the same SCI have operator images with equivalent spectral measures in the sense of theorem 6.8. The physical

quantities from which SCI is computed,-frequencies and decay times,-are the empirical proxies for the spectral measure, not the fundamental objects. SCI compares spectral measures, not physical quantities directly. The commensurability of SCI across domains is not a matter of comparing incommensurable physical quantities but of comparing a single mathematical object,-the spectral measure,-instantiated in different physical systems.

### 7.3 Objections and Responses

**Objection 1: SCI is too coarse to determine unitary equivalence.** A single real number cannot encode the full information required to determine unitary equivalence of operators. Two operators may have identical SCI while having completely different spectral distributions.

*Response.* This is exactly what theorem 6.10 addresses. The conjecture does not assert that SCI determines unitary equivalence for arbitrary operators,-it asserts this for the restricted class $\mathcal{C}_{\mathrm{CS}}$. Whether the class of operators arising from the CS functor construction is sufficiently restricted that SCI is injective on their spectral measures is the key open question. The objection identifies the hardest mathematical problem in the framework, not a reason to abandon it.

**Objection 2: The framework is circular.** If inaccessibility is the criterion of entry into CS, and CS characterises systems by their spectral signatures, then CS characterises systems that are defined by the fact that they cannot be characterised. This appears circular.

*Response.* There is no circularity. Inaccessibility is a scope condition,-it determines which systems CS applies to. Within that scope, CS characterises systems by their morphic structure and spectral signatures, which are accessible even when the internal structure is not. The internal structure of a black hole is inaccessible; its quasinormal mode spectrum is not. CS characterises the accessible spectral properties of systems whose internal properties are inaccessible. These are distinct levels of description and no circularity arises.

**Objection 3: The consciousness application is speculative.** Without a single computed SCI value from real consciousness data, the consciousness application is speculation dressed as theory.

*Response.* This objection conflates two distinct contributions. The theoretical contribution,-establishing that SCI is well-defined for conscious systems, computable from connectome harmonic data, and theoretically superior to existing measures on structural grounds,-does not require empirical computation. The empirical contribution,-validating theorem 5.9 against published data,-is deferred and stated explicitly as deferred. The history of theoretical physics provides ample precedent: general relativity was a theoretical contribution

before its empirical predictions were tested; IIT was a theoretical framework before its empirical programme began.

**Objection 4: CS reduces to existing spectral methods and adds no new mathematics.** The SCI formula $\sum p_n Q_n$ is a weighted sum of quality factors. Quality factors are standard in physics. CS repackages existing tools with categorical language but adds no new mathematical content.

*Response.* The mathematical novelty of CS lies not in any individual component but in four things their combination produces: (i) the functorial grounding,-deriving the spectral computation from the morphic structure of $\mathcal{U}$ rather than imposing it externally, which is new and necessary for the substrate independence claim; (ii) the absolute boundary construction,-grounding the spectral extent in a physically derived operator norm, which is not present in graph spectral or Koopman methods; (iii) the cross-domain comparability established by theorem 3.25, which follows from the functor construction and is not available in domain-specific spectral methods; (iv) theorem 6.4 itself,-the claim that SCI is a complete invariant of categorical equivalence for CS operators,-which is a new mathematical claim with no precedent in the existing spectral literature.

### 7.4 The Open Mathematical Programme

We enumerate all open problems of CS as a unified research programme, organised by dependency structure.

*Foundational problems* (prerequisites for the conjecture): theorem 3.9 (functoriality in general) and theorem 3.11 (self-adjointness).

*Structural problems* (refining the framework): theorem 3.22 (monotonicity of SCI), theorem 3.23 (boundedness of SCI), and theorem 5.10 (directed connectome extension).

*Conjecture-level problems* (the proof programme): theorem 6.7 (characterisation of CS operators) and theorem 6.10 (injectivity of SCI on CS spectral measures).

*Empirical problems* (validation): theorem 5.9 (SCI ordering matches PCI ordering) and theorem 5.11 (SCI,$\Phi$ relationship).

The dependency structure is linear at the foundational level: theorem 3.9 is a prerequisite for theorem 6.7, which is a prerequisite for theorem 6.10, which is the key step in the proof of theorem 6.4.

### 7.5 Scope and Limits

CS applies specifically to systems with inaccessible internal structure in the sense of theorem 3.1. We state explicitly what CS does not claim.

CS does not apply to fully accessible systems. CS does not solve the hard problem of consciousness,-it reframes it as noted in theorem 5.12. CS does not provide a decision procedure for consciousness: the framework provides a mathematical foundation, not automatically a clinical tool. CS does not claim that all physical properties of a system are captured by its spectral signature,-$\sigma(T)$ encodes spectral properties of the operator image of the system's morphic structure, not all physical properties. CS assumes self-adjointness of the operator $T$ throughout; for systems where this assumption fails,-dissipative systems, systems with directed connectivity, systems far from thermodynamic equilibrium,-the framework requires extension.

### 7.6 Relationship to the Broader Mathematical Landscape

The injectivity of SCI on spectral measures,-theorem 6.10,-is related to the moment problem in functional analysis: given a sequence of moments of a measure, when is the measure uniquely determined? SCI is a weighted first moment of the spectral distribution. Whether higher moments are recoverable from SCI,-or whether SCI must be augmented with higher-order functionals to determine the full spectral distribution,-is a question in the theory of moment problems [Akhiezer, 1965]. If the spectral measures of CS operators are uniquely determined by their first moment,-which would follow if $\mathcal{C}_{\mathrm{CS}}$ is sufficiently restricted,-then theorem 6.10 reduces to a classical result in moment theory. This connection provides a concrete mathematical route into the proof of theorem 6.10 and is the most immediate priority for the mathematical programme following the resolution of theorem 3.9.

## 8 Conclusion

This paper has introduced Categorical Spectralism,-a mathematical framework for characterising systems with inaccessible internal structure through their spectral properties rather than their categorical labels. The framework is constructed across three abstraction layers: a category of undefined objects whose identity is exhausted by their internal state transitions; a functor mapping this morphic structure into a complete Hilbert space; and a spectral analysis of the resulting operator whose eigenvalue distribution constitutes the system's spectral signature. From this signature we derive the Spectral Complexity Index,-a scalar functional on the spectrum, invariant under unitary equivalence, directly comparable across physical domains, and computable in polynomial time from empirical spectral data.

The framework has been instantiated in two domains. In the gravitational case, CS identifies the post-merger black hole system GW150914 from its quasinormal mode spectrum without categorical commitment to the nature of the source, yielding SCI(GW150914) $= 3.14 \pm 0.08$ from published LIGO parameter estimates. In the consciousness case, CS proposes SCI as

a theoretically grounded replacement for existing complexity measures,-computationally tractable where $\Phi$ is not, formally derived where PCI is not, and substrate-independent where neither is. The consciousness application is explicitly theoretical; empirical validation via theorem 5.9 is deferred to subsequent work.

The central mathematical claim of the paper,-theorem 6.4,-states that SCI equivalence implies categorical equivalence for systems admitting the CS functor construction, regardless of substrate. The conjecture is precisely stated, its proof target is enumerated as four lemmas in dependency order, and its falsification conditions are made explicit. The critical open problem is theorem 6.10,-the injectivity of SCI on spectral measures within the class of CS operators,-whose resolution is connected to the classical moment problem in functional analysis. The converse direction is already proven: categorical equivalence implies SCI equivalence by the unitary invariance of theorem 3.21. The non-trivial direction remains open.

The mathematical programme of CS is substantial but structured. Its dependencies are explicit, its open problems are precisely stated, and its falsification conditions are concrete. We submit this programme to the mathematical community,-and in particular to the applied category theory, functional analysis, and operator algebra communities,-as an invitation to engagement. The framework stands or falls on the proof or refutation of theorem 6.4. We believe the framework is worth the attempt.

## References


Abbott, B.P. et al. (LIGO Scientific Collaboration) (2016). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116(6), 061102.

Akhiezer, N.I. (1965). *The Classical Moment Problem.* Oliver and Boyd.

Atasoy, S., Donnelly, I. & Pearce, J. (2016). Human brain networks function in connectome-harmonic subspaces. *Nature Communications*, 7, 10340.

Berti, E., Cardoso, V. & Will, C.M. (2006). Gravitational-wave spectroscopy of massive black holes with the space interferometer LISA. *Physical Review D*, 73(6), 064030.

Casali, A.G. et al. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198), 198ra105.

Chalmers, D.J. (1995). Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3), 200,219.

Chung, F.R.K. (1997). *Spectral Graph Theory.* American Mathematical Society.

Connes, A. (1994). *Noncommutative Geometry.* Academic Press.

Conway, J.B. (1990). *A Course in Functional Analysis*, 2nd edition. Springer.

Dirac, P.A.M. (1927). The quantum theory of the emission and absorption of radiation. *Proceedings of the Royal Society A*, 114(767), 243,265.

Döring, A. & Isham, C.J. (2011). "What is a thing?": Topos theory in the foundations of physics. In *New Structures for Physics*, Springer Lecture Notes in Physics, 753, 753,937.

Einstein, A. (1905). Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 17, 891,921.

Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Preussischen Akademie der Wissenschaften*, 844,847.

Eilenberg, S. & Mac Lane, S. (1945). General theory of natural equivalences. *Transactions of the American Mathematical Society*, 58(2), 231,294.

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127,138.

Gelfand, I.M. & Naimark, M.A. (1943). On the imbedding of normed rings into the ring of operators in Hilbert space. *Matematicheskii Sbornik*, 12(2), 197,213.

Halmos, P.R. (1951). *Introduction to Hilbert Space and the Theory of Spectral Multiplicity.* Chelsea Publishing.

Heunen, C. & Landsman, N.P. (2009). Bohrification of operator algebras and quantum logic. *Synthese*, 186(3), 719,752.

Kokkotas, K.D. & Schmidt, B.G. (1999). Quasi-normal modes of stars and black holes. *Living Reviews in Relativity*, 2(1), 2.

Koopman, B.O. (1931). Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5), 315,318.

Ladyman, J. (1998). What is structural realism? *Studies in History and Philosophy of Science*, 29(3), 409,424.

Lurie, J. (2009). *Higher Topos Theory.* Princeton University Press.

Lurie, J. (2018). *Spectral Algebraic Geometry.* Available at https://math.ias.edu/~lurie. Accessed March 2026.

Mac Lane, S. (1971). *Categories for the Working Mathematician.* Springer.

Mezić, I. (2005). Spectral properties of dynamical systems, model reduction and decompositions. *Nonlinear Dynamics*, 41(1,3), 309,325.

Nagel, T. (1974). What is it like to be a bat? *Philosophical Review*, 83(4), 435,450.

Oizumi, M., Albantakis, L. & Tononi, G. (2014). From the phenomenology to the mechanisms of consciousness: Integrated Information Theory 3.0. *PLoS Computational Biology*, 10(5), e1003588.

Paulsen, V. (2002). *Completely Bounded Maps and Operator Algebras.* Cambridge University Press.

Penrose, R. (1965). Gravitational collapse and space-time singularities. *Physical Review Letters*, 14(3), 57,59.

Reed, M. & Simon, B. (1972). *Methods of Modern Mathematical Physics, Vol. 1: Functional Analysis.* Academic Press.

Reed, M. & Simon, B. (1975). *Methods of Modern Mathematical Physics, Vol. 2: Fourier Analysis, Self-Adjointness.* Academic Press.

Rudin, W. (1991). *Functional Analysis*, 2nd edition. McGraw-Hill.

Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(42).

Tononi, G. et al. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450,461.

Tsuchiya, N. et al. (2016). Using category theory to assess the relationship between consciousness and integrated information theory. *Neuroscience Research*, 107, 1,7.

Tsuchiya, N. & Saigo, H. (2021). A relational approach to consciousness: categories of level and contents of consciousness. *Neuroscience of Consciousness*, 2021(2).

Wald, R.M. (1984). *General Relativity.* University of Chicago Press.

Weinberg, S. (1995). *The Quantum Theory of Fields, Vol. 1.* Cambridge University Press.

Worrall, J. (1989). Structural realism: The best of both worlds? *Dialectica*, 43(1,2), 99,124.